

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: lab-2-predictive-analytics

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GitHub Repository Link: [Meh02ajv/lab-2-predictive-analytics](https://github.com/Meh02ajv/lab-2-predictive-analytics)

Date Submitted: Friday, July 03, 2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	25%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

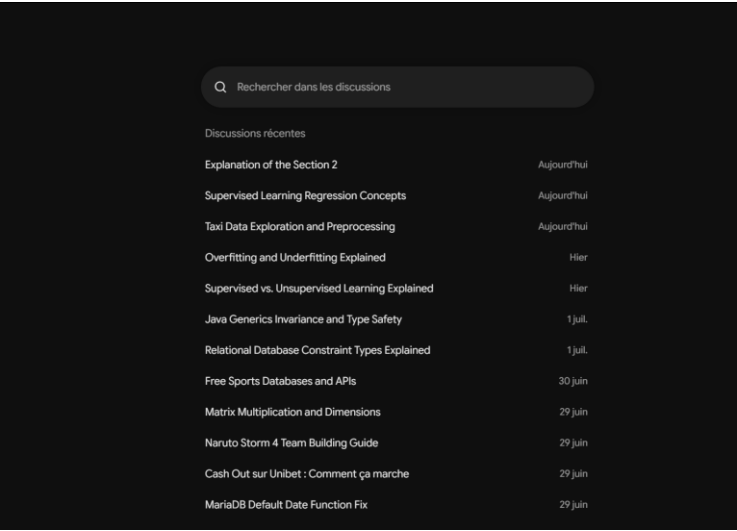
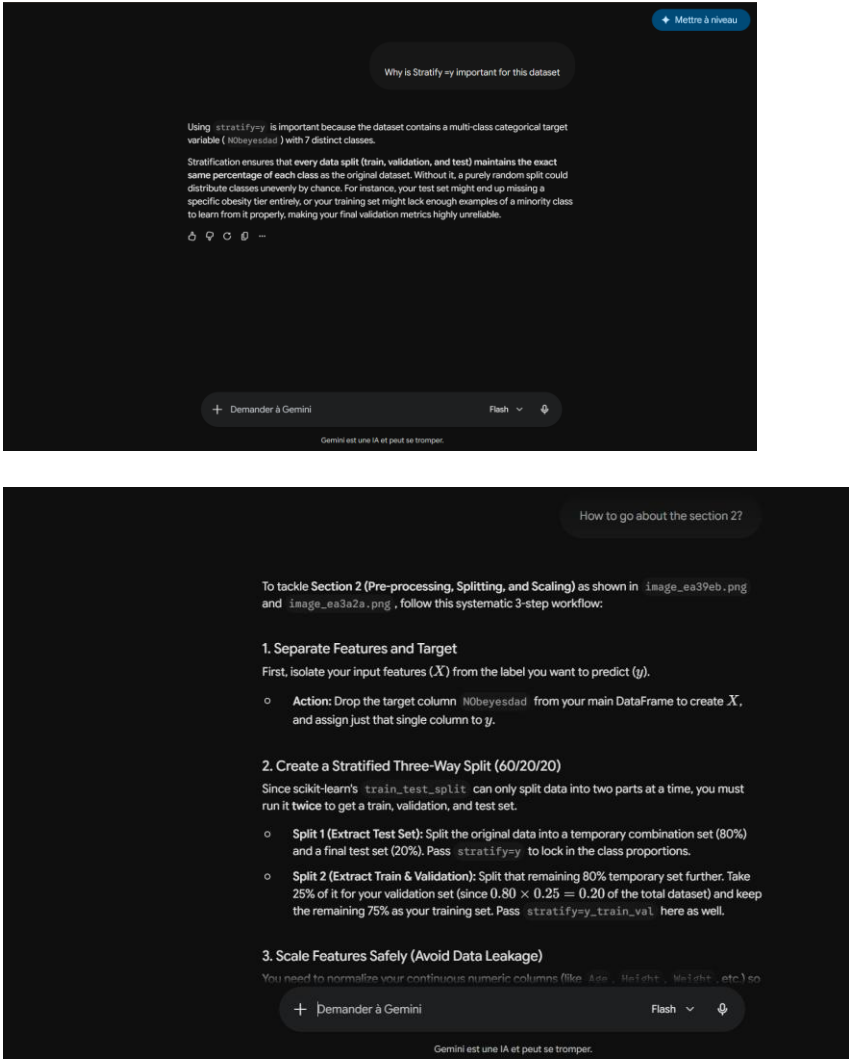
Complete the table below for each AI tool used. Add more rows where necessary.

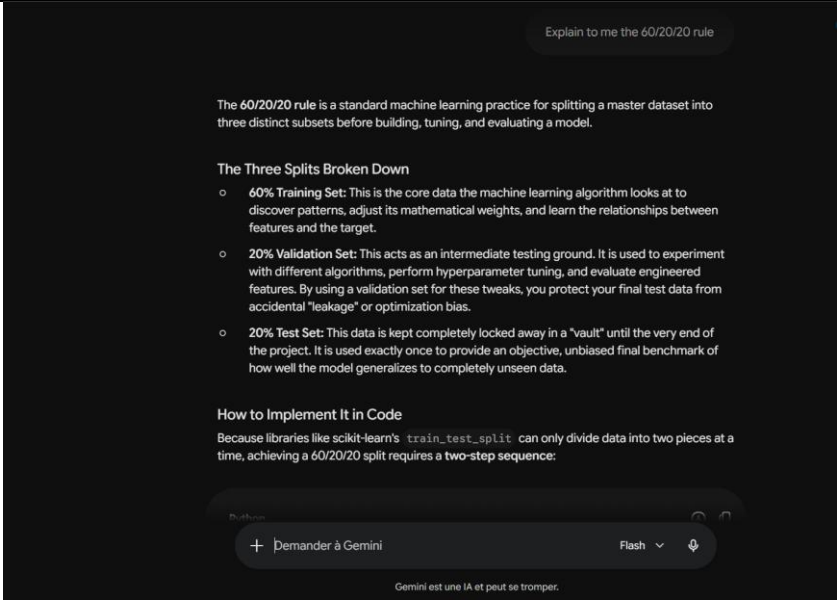
Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Gemini AI	Correction of Errors, Explanation of concepts, Showing the use of syntaxes	- Why is Stratify =y important for this dataset - How to go about that question? - Explain to me the 60/20/20 rule	Part 1.4, 2.2, 2.4, 3.1	By referring to the lab instructions and similar implementations on the internet

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	No	
Prompt history	Yes	

		
Screenshot of AI interaction	Yes	

		 The 60/20/20 rule is a standard machine learning practice for splitting a master dataset into three distinct subsets before building, tuning, and evaluating a model. The Three Splits Broken Down 60% Training Set: This is the core data the machine learning algorithm looks at to discover patterns, adjust its mathematical weights, and learn the relationships between features and the target. 20% Validation Set: This acts as an intermediate testing ground. It is used to experiment with different algorithms, perform hyperparameter tuning, and evaluate engineered features. By using a validation set for these tweaks, you protect your final test data from accidental "leakage" or optimization bias. 20% Test Set: This data is kept completely locked away in a "vault" until the very end of the project. It is used exactly once to provide an objective, unbiased final benchmark of how well the model generalizes to completely unseen data. How to Implement It in Code Because libraries like scikit-learn's <code>train_test_split</code> can only divide data into two pieces at a time, achieving a 60/20/20 split requires a two-step sequence: Demander à Gemini Gemini est une IA et peut se tromper.
Exported AI conversation	No	
AI-generated code snippet	Yes	<pre> binary_cols = ['family_history_with_overweight', 'FAVC', 'SMOKE', 'SCC'] for col in binary_cols: obesity_prepped[col] = obesity_prepped[col].map({'yes': 1, 'no': 0}) # Define the logical inherent sequence for eating/drinking frequencies caec_order = ['no', 'Sometimes', 'Frequently', 'Always'] calc_order = ['no', 'Sometimes', 'Frequently', 'Always'] ordinal_enc = OrdinalEncoder(categories=[caec_order, calc_order]) obesity_prepped[['CAEC', 'CALC']] = ordinal_enc.fit_transform(obesity_prepped[['CAEC', 'CALC']]) # Nominal columns -> pd.get_dummies (One-Hot Encoding) nominal_cols = ['Gender', 'MTRANS'] obesity_prepped = pd.get_dummies(obesity_prepped, columns=nominal_cols, drop_first=True) </pre> <pre> def evaluate_classifier(model, name): print(f"=== {name} Performance ===") sets = {'Train': (X_train, y_train), 'Validation': (X_val, y_val), 'Test': (X_test, y_test)} </pre>

		<pre> for set_name, (X_data, y_true) in sets.items(): y_pred = model.predict(X_data) # Calculate accuracy and macro-F1 acc = accuracy_score(y_true, y_pred) macro_f1 = f1_score(y_true, y_pred, average='macro') print(f'{set_name} Set -> Accuracy: {acc:.4f} Macro F1: {macro_f1:.4f}') print("\n") evaluate_classifier(lr_clf, "Logistic Regression") evaluate_classifier(rf_clf, "Random Forest Classifier") </pre>
Revised version showing student input	No	

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: Hariel

Date: Friday, July 03, 2026